GlucoTwin — Predictive Metabolic Digital Twin

System Architecture & Component Design | CRP Digital Twin Challenge 2026

DATA INGESTION	PREPROCESSING	FEATURE ENGINEERING	ML ENGINE	SIMULATOR
HUPA-UCM Dataset DOI: 10.17632/3hbcscwz44.1 25 T1D patients Abbott FreeStyle Libre 2 CGM CSII + MDI Insulin Pumps Fitbit Ionic Wristbands	5-min CGM Interpolation Leakage-Safe Rolling Windows (closed='left' guard) Forward Label Generation (T → T+60 min) Timestamp Temporal Sort Zero Data Leakage Guarantee	50-Feature Stack CGM Rolling Stats (mean/std/min) Glucose Velocity (15/30 min) IOB & COB Approximations Circadian Phase Vectors (sin/cos) Expanding Baseline Mean/EWM Nocturnal Context Flags	LightGBM Primary Classifier + RandomForest Ensemble Isotonic Calibrator (Brier: 0.069 → 0.039) LOPO Cross-Validation 24 patients | AUC 0.970±0.027 Dynamic Fallback Gate	MetabolicSimulator v2 Damped-Velocity ODE (k=0.015 min■¹) 2-Compartment Bergman Model Rescue Carb Scenarios 15g fast / 25g snack Custom bolus & exercise

▼ TWIN CORE ▼

ALERT STATE MACHINE	PATIENT MEMORY	OOD DETECTOR	TwinState v6 API
3-State FSM: NORMAL→WATCH→ALARM Entry threshold: 0.13 (calibrated) Exit threshold: 0.08 (hysteresis) 15-min cooldown buffer 4.39 alert episodes/day vs 29.21 raw point alarms/day	Persistent per-patient episode log Strict backward-looking guard (< T) Similar episode retrieval (Top-3) Alert transition timeline Baseline mean deviation tracking Expanding + EWM personalisation	Mahalanobis feature-space distance CGM gap continuity check Confidence score: 0.0–1.0 OOD flag + component breakdown Flagged: 0.00% on HUPA0023P Training distribution boundary	Multi-Horizon: 30/60/90/120 min SHAP top-K risk drivers What-If simulation payloads Model self-awareness block JSON-serializable full snapshot TwinState contract v6 sealed

LOPO ROC-AUC 0.970 ± 0.027 (24 patients)	**TTT MAE 32.55 min (11,471 pairs)**	**Brier Reduction 0.069 → 0.039 (−43.5%)**	**ECE Reduction 0.1006 → 0.0055 (−94.5%)**	**Episode Detection 300/300 (100%) Retrospective**	**Lead Time 47.4 min mean 48.8 min median**